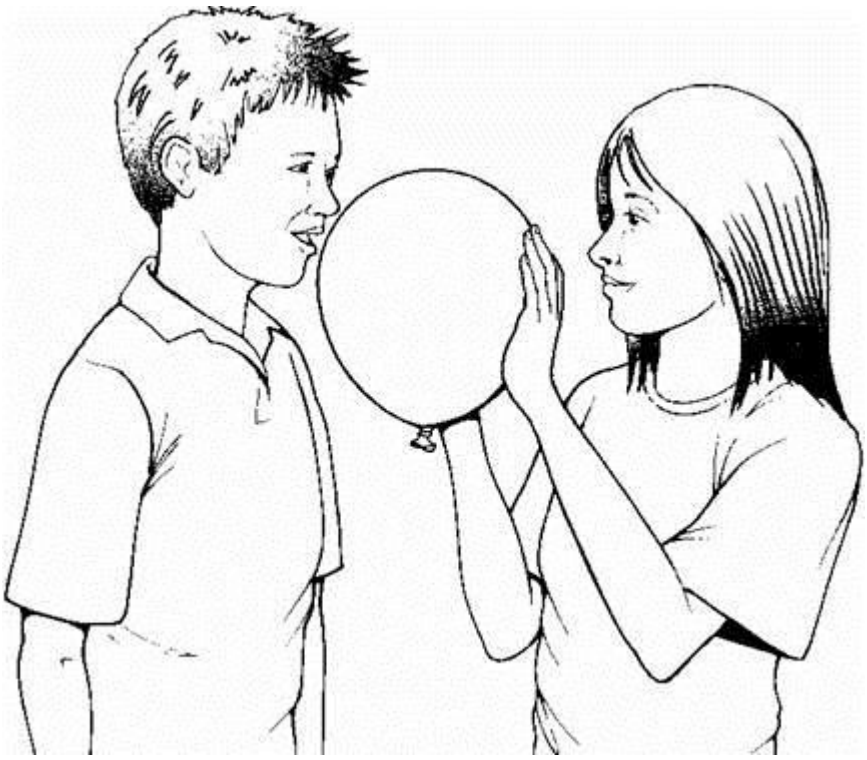
Q1.

Jasmine is deaf. She blows up a balloon and holds it near to John's mouth. She cannot hear John's voice, but she can tell that he is speaking, by feeling the balloon.



John Jasmine

(a) How can Jasmine tell when John is speaking, by feeling the balloon?

.....

.....

1 mark

(b) John shouts loudly. How will the balloon feel different to Jasmine now?

.....

.....

1 mark

(c) Loudness is measured in decibels. The table below shows the loudness of some sounds.

sound	loudness, in decibels
whispering	20
normal talking	60
disco	100
road drill	120
space rocket taking off	190

Jasmine's balloon bursts.
What would be the most likely range of loudness of the sound produced when the balloon bursts?
Tick the correct box.

below 60 decibels ☐

60–120 decibels ☐

120–190 decibels ☐

above 190 decibels ☐

1 mark

(d) (i) Very loud sounds can damage a person's ears.
In what way can the ears be damaged?

.....

.....

1 mark

(ii) Some people work in very noisy places.
How can they protect their ears?

.....

.....

1 mark
Maximum 5 marks

Q1.

(a) the balloon vibrates

accept 'by vibrations'

accept 'it moves'

1 (L4)

(b) it vibrates more strongly

accept 'bigger vibrations'

accept 'it moves more'

accept 'it vibrates more'

*do **not** accept 'more vibrations'*

1 (L4)

(c) 60-120 decibels ✓

if more than one box is ticked, award no mark

1 (L4)

(d) (i) burst ear drum

*accept 'the cochlea or nerve endings **or** inner ear **or** middle ear is damaged'*

accept 'it makes you deaf'

*accept 'tinnitus' **or** 'ringing in the ear'*

1 (L4)

(ii) any **one** from

- wear ear defenders
- use earplugs

accept any appropriate alternative method

1 (L3)

[5]